

SAMIUL HOSSAIN SAJAL

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EDUCATION

(Oct 2022 - Present)

M.Sc. in Physics

University of Cologne, Germany.

Primary: Condensed Matter Physics

Secondary: Computational Physics

(Jun 2019 - Nov 2021)

M.Sc. in Physics
SUST, Bangladesh.

(Mar 2015 - Apr 2019)

B.Sc. in Physics
SUST, Bangladesh

LANGUAGES

- English — Fluent (C1)
- German — A1.2 (currently attending a German course)
- Bangla — Native

SUMMARY

M.Sc. Physics candidate specializing in condensed matter and computational physics, with research experience in TEM image segmentation, grain tracking, and molecular-dynamics simulations. Currently developing computer-vision workflows for quantitative materials analysis at Forschungszentrum Jülich (IAS-9).

RESEARCH EXPERIENCE

Student Research Assistant (Nov 2025 - Present)
IAS-9, Forschungszentrum Jülich

- Developing a computer-vision pipeline for grain-growth analysis in in-situ TEM videos
- Segmenting grains using U-Net, SAM2, SAM3, and MatSAM-based workflows
- Tracking individual grains using overlap analysis and Hungarian matching
- Refining segmentation topology and validating results against manual annotations

Master's Thesis Research (Oct 2019 - Nov 2021)
Shahjalal University of Science & Technology (SUST)
Thesis DOI: 10.13140/RG.2.2.21838.43845

- Studied Al nanoparticle formation and oxidation during pulsed laser ablation in liquids
- Ran ReaxFF/LAMMPS molecular-dynamics simulations on HPC systems
- Analysed temperature-dependent growth, size, and structure using OVITO, VMD, and Python

SKILLS

Research Methods
Computer vision, image and video segmentation, multi-object tracking, model validation, molecular dynamics, EBSD analysis

Machine-Learning Models
U-Net, SAM, SAM2, SAM3, MatSAM, prompt-based segmentation

Scientific Computing
Python, PyTorch, OpenCV, scikit-image, NumPy, SciPy, Matplotlib

Simulation & Research Tools
LAMMPS, ReaxFF, OVITO, VMD, Napari, Linux, HPC, Git, LaTeX

SELECTED PRESENTATIONS

Poster Presentation: Atomic-scale investigation of the formation of Al nanoclusters in the Pulsed Laser Ablation Technique. (*Bangladesh Physical Society, 2021)*

SELECTED COURSEWORK & TRAINING

Deep Learning and Physics – *University of Cologne* (Feb 2026)
Deep Learning with PyTorch: Image Segmentation – *Coursera Project* (Oct 2025)
Computational Many-Body Physics – *University of Cologne* (Feb 2025)
Scientific Programming with Python – *University of Cologne* (Jul 2023)
Google Data Analytics Professional Certificate – *Coursera* (Sep 2022)

ADDITIONAL WORK EXPERIENCE

Restaurant Employee - *Atawich, ATACO GmbH* (Jan 2024 - Sep 2025)
Sortation Associate - *Amazon Deutschland Service GmbH* (Jun 2023 - Sep 2023)